

Predicting Neighborhood Transit Recovery

Analyzing NYC Subway Ridership Patterns & Economic Factors Post-COVID

CSE 6242 – DATA AND VISUAL ANALYTICS

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What Are We Trying to Do?

HEILMEIER QUESTION #1

We aim to identify which NYC neighborhoods recovered fastest from COVID-19's impact on subway ridership and determine what economic and demographic factors predict faster recovery.

Scale of Analysis

262 Neighborhood Tabulation Areas (NTAs) across New York City, covering January 2020 through December 2023, a full four-year window spanning the pandemic and its aftermath.

The Core Distinction

"Bouncing back" from COVID lows and **"truly recovering"** to pre-pandemic levels are fundamentally different phenomena and conflating them leads to misleading conclusions about urban resilience.

262

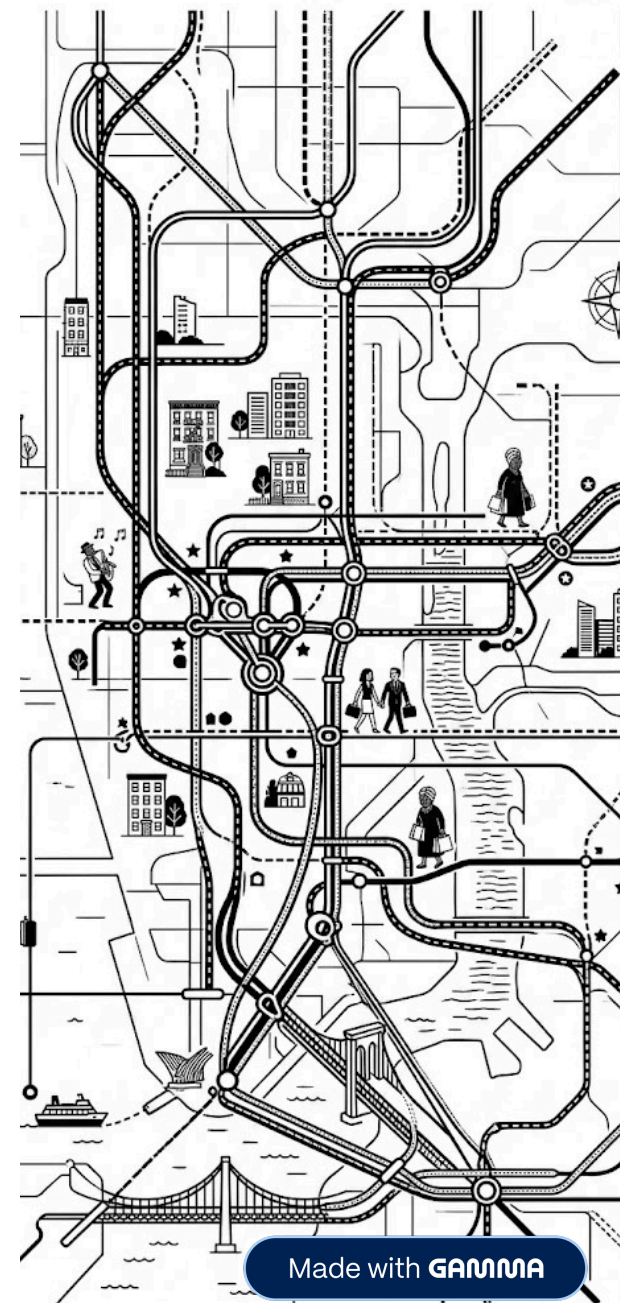
NTAs Analyzed

Neighborhood Tabulation Areas across all five boroughs

4yr

Time Window

Jan 2020 – Dec 2023 full coverage



How Is It Done Today & What Are the Limits?

HEILMEIER QUESTION #2

Current approaches leave critical gaps in understanding neighborhood-level transit recovery. Existing research is either too aggregate or too narrow in scope.

MTA Aggregate Stats

Publishes system-wide ridership data but provides **no neighborhood-level recovery analysis**, making it impossible to identify which communities were left behind.

Liu et al. (2020)

Documented a **100-year low** in US transit demand during COVID, but analysis remained aggregate only, with no geographic disaggregation.

Ziedan et al. (2023)

Identified **June 2021** as the national recovery turning point, but captured no neighborhood variation beneath the city-wide trend.

Mahfouz et al. (2024)

Examined NYC recovery but focused on **taxis and rideshare**, missing the critical subway-economy linkage that defines neighborhood resilience.

- ❏ **The Gap:** No existing tool integrates transit recovery with neighborhood economic indicators in an interactive, neighborhood-level framework.

What Is New in Our Approach?

HEILMEIER QUESTION #3

Three key methodological innovations set this project apart from all prior work, plus a fully interactive visualization dashboard.

1

Dual Recovery Metrics

Recovery Index measures bounce-back from COVID low. **True Recovery Index** measures return to pre-pandemic baseline. These capture fundamentally different phenomena.

2

Trajectory Clustering

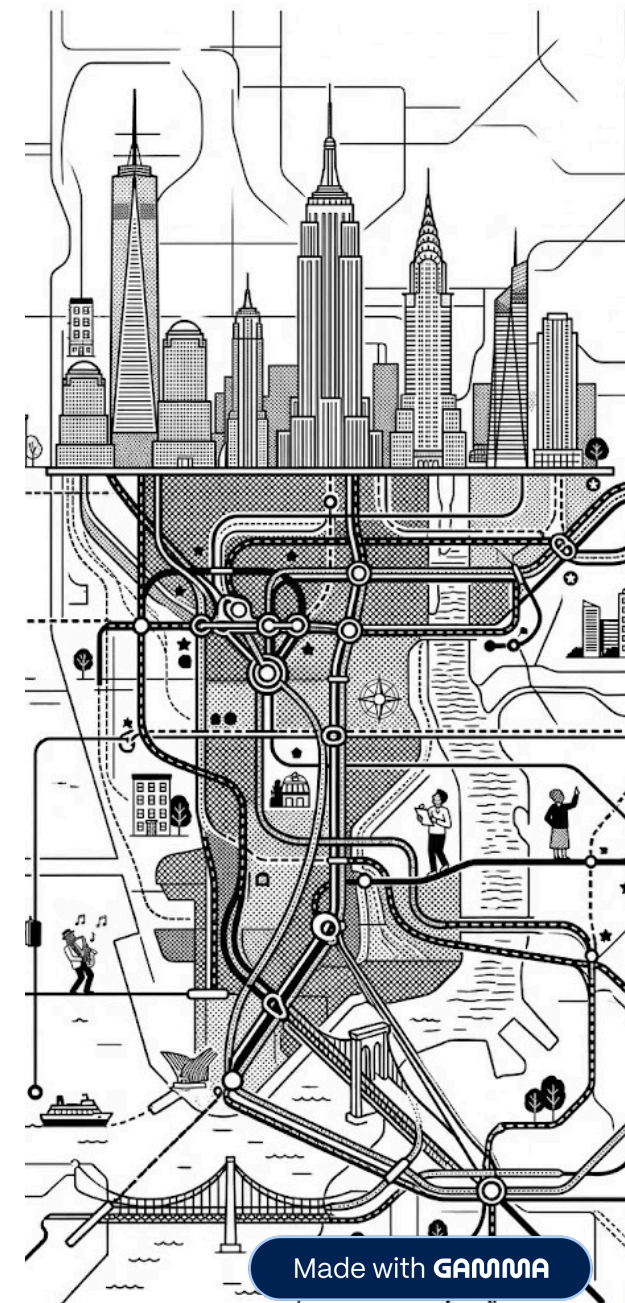
K-means clustering groups all 262 neighborhoods into distinct recovery pattern archetypes — revealing structural differences invisible in aggregate statistics.

3

Remote Work Scoring

Classify **4.2 million jobs** by NAICS industry code into high, medium, and low remote work potential, linking employment structure to transit recovery speed.

Plus an **interactive scroll-snap dashboard** featuring choropleth maps, animated timelines, and drill-down tooltips for neighborhood-level exploration.



Who Cares?

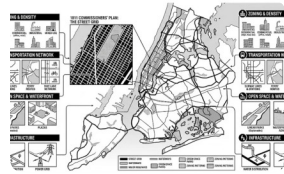
HEILMEIER QUESTION #4

This research serves a broad coalition of stakeholders, from transit operators to policymakers, all of whom need neighborhood-level insight to make better decisions.



MTA: 3.5M Daily Riders

Data-driven service planning and resource allocation across the five boroughs. Knowing which neighborhoods lag enables targeted frequency and capacity adjustments.



Urban Planners

Evidence-based framework for equitable recovery investment. Identifies underserved neighborhoods that need infrastructure and economic development support.



Researchers

Neighborhood-level urban resilience analysis methodology, moving beyond city-wide aggregates to reveal the heterogeneity hidden inside average statistics.



Policymakers

Understanding which communities were left behind in the recovery, and what structural factors explain the disparity to design more targeted interventions.

If Successful, What Difference Will It Make?

HEILMEIER QUESTION #5

→ Actionable MTA Insights

Neighborhood-level service planning recommendations grounded in empirical recovery trajectories, not system-wide averages.

→ Publishable Methodology

A replicable dual-index framework for urban resilience research applicable to any city with open transit and economic data.

→ Reusable Interactive Tool

An open-source dashboard for ongoing NYC neighborhood analysis, extensible beyond the COVID recovery context.

72%

True Recovery

Actual return to pre-pandemic baseline — far lower than commonly reported

2.4x

Bounce-Back

Apparent recovery from COVID low — a misleading metric on its own

📄 **Key Finding:** Standard "recovery" metrics are misleading. True recovery sits at only 72% while the bounce-back metric reads 2.4x, a critical distinction for policy.

Literature Survey

5+ PEER-REVIEWED SOURCES

COVID Transit Impact

Liu, Miller & Scheff (2020)

US transit demand fell to 100-year lows.
PLOS ONE.

Ziedan, Brakewood & Watkins (2023)

Nationwide recovery inflection points identified. *J. Public Transportation*.

Srinivasan et al. (2025)

Essential worker concentration predicted bus retention in Boston.
Transportation Research Part A.

Economic Recovery

Che, Lee & Kim (2023)

Neighborhood retail clusters recovered faster than district-level aggregates.
Cities.

Forouhar et al. (2025)

76% average downtown recovery across 66 North American cities. *Urban Studies*.

Transit–Economy Linkage

Osorio, Liu & Ouyang (2022)

Remote work explained **66%** of Chicago rail ridership loss — the strongest single predictor identified in the literature.
Transportation Research Part D.

📄 This finding directly motivates our remote work scoring methodology applied to 4.2M NYC jobs.

Data & Methodology

FOUR INTEGRATED PUBLIC DATASETS



MTA Subway Hourly Ridership

270M+ rides from data.ny.gov — the primary outcome variable for all recovery index calculations.



PLUTO Property Data

857K tax lots providing land use characteristics — commercial density, residential mix, zoning classifications.



LEHD Employment (WAC)

4.2M jobs classified by NAICS code for remote work potential scoring — high, medium, and low categories.



Census ACS Demographics

2,327 tracts providing income, education, and race data for equity analysis across neighborhoods.

Analytical Methods

01

OLS Regression

With VIF checks to detect multicollinearity among economic predictors

02

5-Fold Cross-Validation

Ensuring model generalizability across held-out neighborhood subsets

03

K-Means Clustering (k=4)

Grouping 262 NTAs into distinct recovery trajectory archetypes

04

Moran's I Spatial Autocorrelation

Testing whether recovery patterns cluster geographically across boroughs

Risks & Mitigations

HEILMEIER QUESTIONS #6-7

Risk Assessment & Mitigations

Weak Predictive Signal

If R^2 is modest, insights remain valuable. Mitigation: multiple model specifications, feature engineering, and ensemble approaches. Even partial explanatory power advances the literature.

Data Quality Issues

Public datasets may have gaps or inconsistencies. Mitigation: cross-validate across multiple sources, document all limitations transparently in the final report.

Visualization Complexity

Interactive dashboards can become unwieldy. Mitigation: use established libraries (Plotly, Folium) with iterative user testing and progressive disclosure design patterns.

Total Project Cost

\$0

Budget Required

All data freely available from public sources. Computing on personal machines and free cloud tiers. Software stack is entirely open source.

Data

Free public sources

Compute

Personal + free cloud

Software

Python, Plotly, Folium

Timeline & Success Criteria

HEILMEIER QUESTIONS #8-9



Midterm Checkpoints

- All four datasets integrated at NTA level
- Dual recovery index calculated for all 262 neighborhoods
- Initial clustering results with k=4 archetypes
- Visualization wireframes complete and reviewed

Final Success Criteria

- **3+ significant predictors** at $p < 0.05$ in regression models
- **Meaningful cluster separation** with interpretable recovery archetypes
- **Interactive dashboard** with choropleth map exploration and neighborhood drill-down